1.0 Purpose

The purpose of this document is to provide product specifications for applicable Hornet Implantable Pulse Generator IPG and charger mechanical aspects.

2.0 Scope

In scope for this document are the product specifications inclusive to the Hornet IPG and charger; as part of the CARSS program Nest 1. The CARSS program stands for Center for Autonomic Nerve Recording and Stimulation Systems. The CARSS is part of the NIH Human Open Research Neural Engineering Technologies (HORNET) initiative for open-architecture, open-source implantable neuromodulation system. This includes a collaborative team between University of Southern California (USC, Los Angeles, CA), Medipace Inc (Pasedena, CA), and Med-Ally, LLC (Goose Creek, SC). The purpose of the system which the IPG and charger is a portion of is to enable community-supported neurotechnology platform to remove technical and financial hurdles to getting access to implantable neuromodulation technologies. Nest 1 focuses on the IPG and charger mechanical aspects.

3.0 Definitions

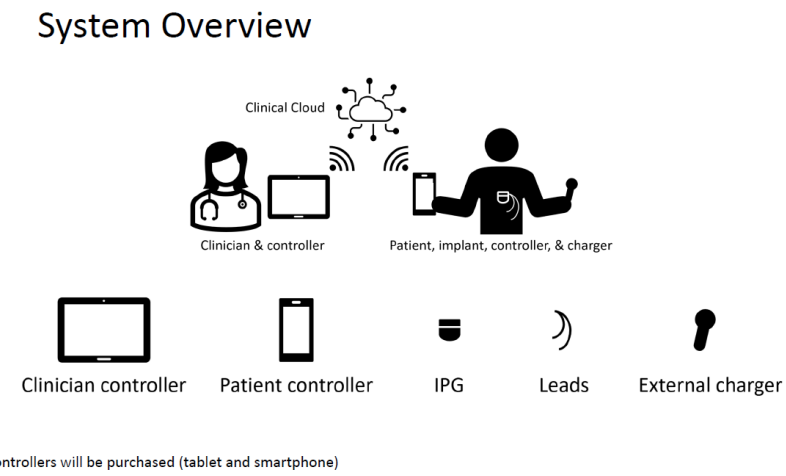
Abbreviation	Definition
BLE	Bluetooth Low Energy. Operates on frequencies: 2.400 GHz – 2.4835 GHz
Controller	A computing device capable of running the NBS-EPG and IPG Controller application(s) tailored for physicians, the patient, or researchers.
IPG	Implantable Pulse Generator
HW	Hardware
SW	Software
PCB	Printed Circuit Board
PSD	Product Specifications Document
GUI	Graphical User Interface
HIPAA	Health Insurance Portability and Accountability Act of 1996. U.S. legislation that provides data privacy and security for safeguarding medical information.
Wi-Fi	A facility allowing an electronic device to wirelessly transfer data or connect to the Internet using IEEE 802.11 standards. Operates on frequencies: 900 MHz, 2.4 GHz, 4.9 GHz, 5 GHz, 5.9 GHz and 60 GHz.
NIH	National Institute for Health
CARRS	Center for Autonomic Nerve Recording and Stimulation Systems
Hornet	Human Open Research Neural Engineering Technologies

Abbreviation	Definition
Nest	Major component or module of the CARRS system

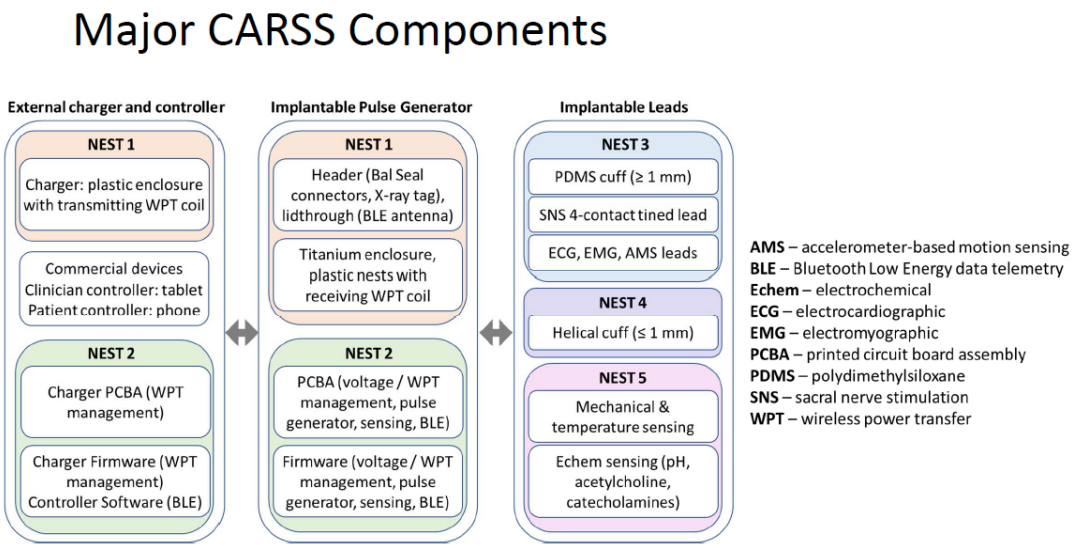
4.0 References

5.0 General Description of Product

The CARRS system is an open source system to enable developments in stimulation and sensing technologies. The system is capable of closed loop stimulation and sensing or can be used independently. The system components are: external charger and controller, implantable pulse generator, collection of interoperable and implantable leads.



Each portion of the CARSS system has been split into major components called nests. See figure below for details of major components or modules of the CARRS system:



This document only addresses Nest 1.

The IPG and charger for the CARRS system will be designed with the following engineering design principles as applicable:

- Material and component selection for maximizing the performance.
- Long-term reliability, redundancy, and fail-safe operations.
- Modularity, plug-and-play operation and off-the-shelf components.
- Preference for simpler and proven solutions over more complex and higher-risk ones.

6.0 Mechanical Specifications

6.1 IPG

PRD ID	IPG Specification Text
PRD-IPG-0001	The IPG incorporates 16 Lead Contacts
PRD-IPG-0002	The contacts are arranged in the header in 4 sets of 4 contacts.
PRD-IPG-0003	With only a small modification, the header can be arranged with one or both lead bores in an 8 contact configuration.
PRD-IPG-0004	The lead connectors are Bal Seal Sygnus 2.8 mm pitch connectors with accompanying silicone seals
PRD-IPG-0005	Bluetooth or WiFi antenna (2.4 GHz capable) is located in the header
PRD-IPG-0006	Hermetic assembly accommodates 2 Quallion QL0200I EnerSys Batteries or similar

PRD-IPG-0007	Hermetic Assembly encloses a charge coil similar to WPT coil 760308101107, Wurth Elektronik
PRD-IPG-0008	Hermetic Assembly encloses the suppled PCBA
PRD-IPG-0009	A nest that supports and stabilizes the PCBA, coil, and batteries within the hermetic assembly
PRD-IPG-0010	The IPG electronics assembly will be hermetically sealed in an enclosure that will be sealed to a helium leak rate no greater than $1X10^{-8}$ in accordance with the MIL STD 883K
PRD-IPG-0011	The IPG shall have a header with a location to secure to the subject with a suture
PRD-IPG-0012	The IPG has a connector block with set screw on each of the 4 (or in the case of an octopolar leads – 2) to mechanically lock in the lead into each lead bore.
PRD-IPG-0013	A set screw is packaged with the IPG to use with the set screw. The torque wrench is 5 in-oz Hex
PRD-IPG-0014	The IPG accommodates 2 EnerSys Quallion QL02001 batteries and if needed can accommodate an off the shelf primary battery greater than 0.75AHr if possible.
PRD-IPG-0015	The IPG outermost dimensions are less than 2” X 2.6” X 0.5”
	IPG Materials
PRD-IPG-2001	All the materials of the IPG Kit that are in contact with the body are selected for known biocompatibility.
PRD-IPG-2002	The IPG case is be made of Grade 23 titanium
PRD-IPG-2003	The IPG header is made of epoxy (EPOTEK 301) to enclose the header components
PRD-IPG-2004	The IPG has a feedthrough consisting of, at minimum, of a titanium lid, ceramic insulation components, and Pt-Ir pins with gold braze
PRD-IPG-2005	The Bal Seals are made of MP35N housing with Pt-Ir springs and incorporates Med-4850 silicone spacer
PRD-IPG-2006	A connector block and set screw are made of 316L
	IPG Testing
PRD-IPG-3001	The IPG electronic assembly should be able to withstand (function after experiencing) the following: Shelf drop per ISO 14708-1:10.1, 23.2, 23.7,
PRD-IPG-3002	The IPG electronic assembly should be able to withstand (function after experiencing) the following: Vibration test at 5-500 Hz for 30 mins per ISO 14708-1:23.2,

PRD-IPG-3003	The IPG electronic assembly should be able to withstand (function after experiencing) the following: Mechanical shock at 500 g per ISO 14708-1:19.1
PRD-IPG-3004	The IPG electronic assembly should be able to withstand (function after experiencing) the following: Accelerated aging per ISO 14708-1:19-1
PRD-IPG-3005	The IPG and Charger should be able to meet heat testing during charging per ISO 14708-1:17.1 & EN 60601-1:11.1.2.2
PRD-IPG-3006	The IPG header will have limited air bubbles per MIL-STD-750-2A (clause 3.1.1.5) where there should be an absence of air bubbles covering more than one-third of the distance from the nearest metal surface to the edge.

6.2 Charger

PRD ID	Charger Specification Text
PRD-CRG-0001	The mating charging coil to the IPG is housed in the charger such as the WPT coil: 760308100110, Wurth Elektronik
PRD-CRG-0002	The charging activities are managed by a PCBA located in the charger
PRD-CRG-0003	The charger is powered by rechargeable or replaceable batteries
PRD-CRG-0004	The charger contains an indicator that provides feedback to the user on the charge status
PRD-CRG-0005	The charger is enclosed in a plastic enclosure
	Charger Testing
PRD-CRG-3001	The charger electronic assembly is supported to be able to withstand (function after experiencing) the following: Shelf drop per ISO 14708-1:10.1, 23.2, 23.7,
PRD-CRG-3002	The charger electronic assembly is supported to be able to withstand (function after experiencing) the following: Vibration test at 5-500 Hz for 30 mins per ISO 14708-1:23.2,
PRD-CRG-3003	The charger electronic assembly is supported to be able to withstand (function after experiencing) the following: Mechanical shock at 500 g per ISO 14708-1:19.1
PRD-CRG-3004	The IPG and Charger should be able to meet heat testing during charging per ISO 14708-1:17.1 & EN 60601-1:11.1.2.2